UVM

1. High-Level UVM Architecture

Test run_test() Environment (env) Agent Sequencer Driver ----> DUT Monitor Scoreboard

2. Key UVM Components

2.1 Test

- Top-level controller
- Starts simulation using run_test()
- Configures environment

2.2 Environment (env)

- Contains agent and scoreboard
- Connects components together

2.3 Agent

- Groups sequencer, driver, monitor
- Represents one interface (I2C bus here)

2.4 Sequencer

- Generates transactions (sequence items)
- Sends them to driver

Example transaction: - Address = 0x50 - Data = 0xA5, 0x5A

2.5 Driver

- Converts transactions into signal-level activity
- Uses interface (i2c_if)

Ex: - Drives SDA and SCL - Creates START, DATA, STOP

2.6 Monitor

- Passively observes signals
- Reconstructs transactions
- Does NOT drive signals

2.7 Scoreboard

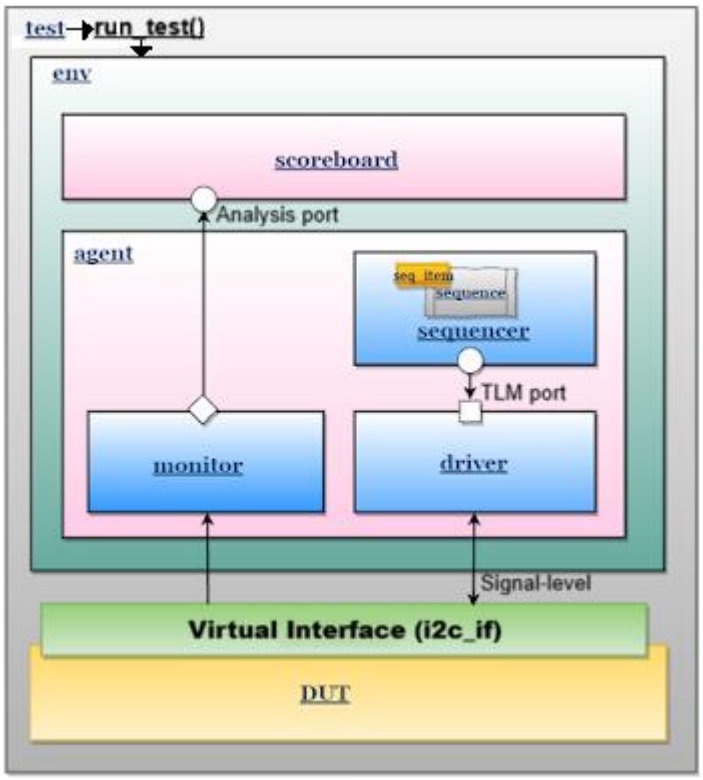
- Compares expected vs actual
- Reports PASS or FAIL

3. Transaction-Level Modeling (TLM)

UVM uses TLM for communication between components.

Connections:

- Sequencer - Driver - seq_item_port



- Monitor - Scoreboard - analysis_port

4. UVM Phases

Build Phase

- Create components

Connect Phase

- Connect TLM ports

Run Phase

- Actual simulation happens

Extract Phase

- Collect data

Report Phase

- Print results

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5. Flow - Project

Step 1: Test Starts

run_test() is called

Step 2: Sequence Created

Transaction: - Address = 0x50 - Data = 0xA5, 0x5A

Step 3: Sequencer Sends Transaction

Transaction sent to driver

Step 4: Driver Executes

Driver uses interface functions:

- master_drive_sda_low()
- master_release_sda()

Creates actual waveform

Step 5: DUT Processes

I2C master FSM runs

Step 6: Monitor Observes

- Reads SDA and SCL
- Reconstructs transaction

Step 7: Scoreboard Checks

Compares: - Expected data - Observed data

Result PASS

6. Code Level Understanding

6.1 Sequence Item

Represents transaction:

- Address
- Data
- Direction

6.2 Driver Logic

Driver converts transaction to signals:

Example:

- Generate START

- Send address
- Wait for ACK
- Send data

6.3 Monitor Logic

- Samples SDA and SCL
- Detects START and STOP
- Extracts data bits

6.4 Scoreboard Logic

- Receives transaction from monitor
- Compares with expected

7. Example Execution

Transaction:

- Address = 0x50
- Data = 0xA5, 0x5A Flow:

START - ADDRESS - ACK - DATA - ACK - DATA - ACK - STOP

8. Simulation Output

UVM Log shows:

- Driver sending data
- Slave responding
- Monitor observing

- Scoreboard PASS

```
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :   13
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [DRV]     1
# [MON]     1
# [Questa UVM]     2
# [RNTST]     1
# [SCB]     2
# [SLV]     4
# [TEST_DONE]     1
# [UVM/RELNOTES]     1
#
# ** Note: $finish    : /usr/share/questa/questasim/verilog_src/uvm-1.2/src/base/uvm_root.svh(517)
#    Time: 805 ns  Iteration: 69  Instance: /tb_top
# End time: 12:50:07 on Mar 28,2026, Elapsed time: 0:00:03
# Errors: 0, Warnings: 0
End time: 12:50:07 on Mar 28,2026, Elapsed time: 0:00:17
*** Summary ********************************************
    qrun: Errors:   0, Warnings:   0
    vlog: Errors:   0, Warnings:   0
    vopt: Errors:   0, Warnings:   0
    vsim: Errors:   0, Warnings:   0
  Totals: Errors:   0, Warnings:   0
Finding VCD file...
./dump.vcd
[2026-03-28 16:50:08 UTC] Opening EPWave...
Done
```